

Network Types -- Guided Notes ANSWER KEY

Unit tp-2.9 | CompTIA Network+ N10-009 Obj. FC0-U71 2.9 | N10-008 FC0-U71 2.9

For instructor use / distribute after students complete the notes.

Question 1

A LAN (Local Area Network) covers a _____ geographic area such as a single _____ or office. Devices on a LAN communicate at _____ speeds (typically 1 Gbps on modern wired networks). The key device connecting LAN devices is the _____.

Answer: Small; home (building); high; switch.

Question 2

A WAN (Wide Area Network) spans _____ geographic areas and connects multiple LANs. The Internet is the largest example of a _____. WANs use _____ provided by ISPs (leased lines, fiber, etc.) to link distant locations. WAN links typically have _____ (higher/lower) latency than LAN.

Answer: Large; WAN; infrastructure (carrier links); higher.

Question 3

A WLAN (Wireless LAN) uses _____ signals instead of physical cables to connect devices. The IEEE standard for Wi-Fi is _____. A WLAN is still a _____ -- it just uses wireless transmission within that local area. A _____ provides the wireless signal in a WLAN.

Answer: Radio; 802.11; LAN; WAP (wireless access point).

Question 4

A MAN (Metropolitan Area Network) is larger than a _____ but smaller than a _____. It typically covers a _____, campus, or city district. Examples include a city-wide cable TV network or a _____ network connecting multiple university buildings.

Answer: LAN; WAN; city (metropolitan area); municipal.

Question 5

A PAN (Personal Area Network) covers a very _____ range (typically _____ meters) and connects personal devices near a single individual. _____ and _____ are the most common PAN technologies. An example is a laptop connected to wireless earbuds.

Answer: Short; 10; Bluetooth; NFC (near-field communication).

Question 6

A VPN (Virtual Private Network) creates an _____ tunnel over a public network (like the Internet), allowing remote users to access a _____ network as if they were physically present on it. Data inside the VPN tunnel is _____, protecting it from interception on the public network.

Answer: Encrypted; private (corporate); encrypted.

Question 7

An intranet is a _____ network accessible only to authorized internal users within an organization. An extranet extends access to _____ (partners, vendors) with appropriate permissions. Neither is the same as the _____, which is a public network accessible to anyone.

Answer: Private (internal); external parties; Internet.

Question 8 -- Real World Application

REAL WORLD: A company has offices in Austin, TX and Boston, MA. Each office has its own LAN. Developers in both offices need to access the same internal code repository server located in Austin. Describe the network types involved and explain how a VPN could be used to give the Boston team secure access to the Austin server across the Internet.

Answer: Each office has a local LAN for internal device communication. The two LANs are connected across the Internet (a WAN). The company can use a site-to-site VPN by installing VPN concentrators at each office. The concentrators establish an encrypted tunnel over the Internet between Austin and Boston. All traffic between the two LANs travels through this tunnel. The Boston developers can reach the Austin code repository server as if they were on the local Austin network, and all data in transit is encrypted. No external attacker intercepting the Internet traffic can read the contents because it appears as encrypted VPN packets. This is more secure than exposing the code repository directly to the public Internet.